The Linear Regression Model

Lecture 1 (R code)

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Reading

This lecture covers material from Wooldridge (2025):

- Chapter 2.1 - "Definition of the Simple Regression Model"
- Chapter 2.7* - "Regression on a Binary Explanatory Variable"
- Chapter 3.1 - "Motivation for Multiple Regression"

*We will cover the Potential Outcomes Framework (Ch. 2.7a - "Counterfactual (or Potential) Outcomes, Causality, and Policy Analysis") in Week 7. For now, it is only important to know the interpretation of a regression on a binary ("dummy") variable.

Linear Regression Model

General form:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

Component #1: Data

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

The **observable** variables in the model are,

- y: the outcome (dependent) variable,
- $x_1, x_2, \ldots, x_k$: the k regressors (independent variables).[1]

Component #2: Parameters

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

The relationship between the outcome and regressors is explained by set of *non-random* population parameters:

- $\beta_1, \beta_2, ..., \beta_k$ are often referred to as "slope coefficients",
- β_0 is the "intercept".

> **!** Linear in Parameters
>
> The model is referred to as the *linear* regression model because it is linear in parameters, NOT because it is linear in regressors. The x_j are generic placeholder names. Consider, $x_2 = x_1^2$, the model would be non-linear in x_1. However, it would remain linear in parameters.

Component #3: Error

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

The error term represents the part of y that is NOT explained by the *regression function*.[2]

In Econometrics, a lot of time is spent thinking about the error term:

- Theoretically, we will want to assume that it can be 'ignored'
- Practically, this often too strong an assumption to make.

[1] There are other words given to the righthand-side variables; including "control variables" and "covariates". These terms have a particular meaning which will become clearer at a later stage.

[2] It is inaccurate to say "NOT explained by the regressors", as the the error term may contain non-linear functions of the regressors.

Vector notation

When reading papers, you will frequently see the following notation:

$$y = \mathbf{x}\beta + u$$

What's going on here? The variables have been collected as vectors:

$$y = \underbrace{\begin{bmatrix} 1 & x_1 & x_2 & \dots & x_k \end{bmatrix}}_{\mathbf{x}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix}}_{\beta} + u$$

When written in vector notation $\mathbf{x}$ always precedes β, as $\beta\mathbf{x} \neq \mathbf{x}\beta$.

> **Vector notation differences**
>
> Wooldridge defines $\mathbf{x}$ as a row vector, but β as a column vector. Many texts define both as column vectors and will therefore express the linear population regression function as,
>
> $$y = \mathbf{x}'\beta + u$$

Sample

When we have a sample of observations, $i = 1, \dots, n$, we use the notation:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + u_i$$

We will need to make further assumptions about sampling,

- do observations relate to one another?
- are they derived from the same population model?

> **Unit of Analysis**
>
> An often overlooked feature of the model is the unit of analysis. In the model, the subscript i represents the unit at which the data is observed. For example, one model may represent data at the individual level, while another captures firm-level data. When each observation represents the same unit observed across different time periods, we usually

use the notation t: $\{y_t, \mathbf{x}_t\} : \ t = 1, \ldots, T$. In panel or repeated cross-section datasets, the data is observed at two levels: i and t.

How did we get here?

Let's take a step back and ask how we got here. Where does the Linear Population Regression Function come from? This is actually not an easy question to answer and there is more than one way to arrive at a model of this form.

Approach #1: DGP

The model represents the true **data generating process** at the population level $\Rightarrow$

- if we know the population parameters
- AND the joint distribution of the regressors and error term

we can generate data in a manner akin to sampling from the population distribution.

e.g.

$$y_i = \beta_0 + \beta_1 x_i + u_i$$
$$\text{with } \beta_0 = 0.5, \ \beta_1 = 0.6, \ x_i \sim U(0,1), \ u_i \sim N(0, 0.6^2)$$

Population

When econometricians uses the word 'population', they are not referring to an enumerable collection of units, as in a census. They mean a probability distribution — the joint distribution over the random variables of interest, from which the observed sample is drawn. As we shall shortly see, the population parameters are defined in terms of the joint distribution of the random variables.

Finite vs Super Population

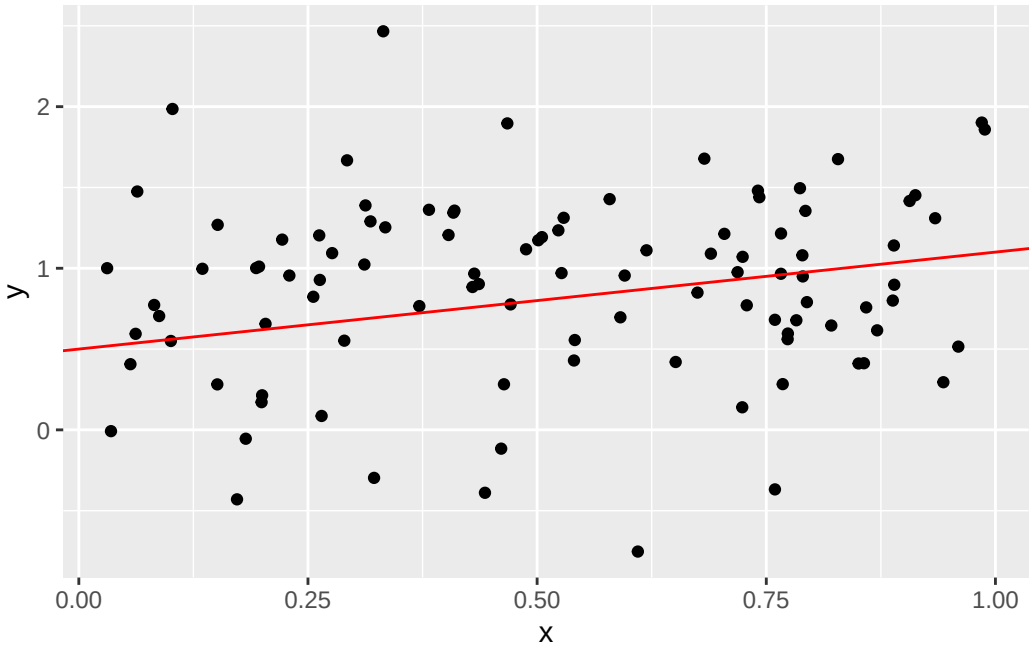
In design-based causal inference methods, a distinction is made bewteen the 'finite' and 'super' populations. The above description of the population fits the super-population approach: each sample is a draw from a theoretically infinite super-population. This sampling uncertainty is key to understanding the properties of the estimator. In the finite-population approach, the sample is treated as the population and uncertainty arises *only* from random assignment in an experiment.

```r
# Parameters
n   <- 100
b0 <- 0.5
b1 <- 0.6
sd_norm <- 0.6

# Variables
x     <- runif(n)
error <- rnorm(n,sd=sd_norm)
y     <- b0 + b1 * x + error

df <- data.frame(x, y)

# Plot
ggplot(df, aes(x, y)) +
  geom_point() +
  geom_abline(intercept = b0, slope = b1, color = "red")
```



Econometricians use such approaches to study the properties of estimators. It is often referred to as a Monte Carlo simulation. Given that you know the true value of the population parameters, you can examine how well the estimator performs across multiple simulations.

Approach #2: Prediction

Suppose you want to predict y using information from a set of variables $\mathbf{x} = [x_1, x_2, \dots, x_k]$. The 'best' predictor of y is the conditional expectation (function; CEF)

$$E[y|\mathbf{x}] = m(\mathbf{x})$$

Where the CEF is a function of $\mathbf{x}$.[3]

This is the starting point for Data Science.

- 'Fancier' methods like Lasso, Random Forests, and Neural Networks are simply newer methods to estimate $m(\mathbf{x})$. They have core advantages over linear regression models that relate specifically to 'big' data.

> **Best? Big?**
>
> What do we mean by *best* predictor? This is a very specific use of the word 'best'. The CEF is the best mean-squared error predictor of y. You can find more on this in the additional material on the CEF.
> 'Big' data also has a very specific meaning. It does not concern the absolute size (i.e. number of observations n) of the dataset, but rather the number of regressors/predictors (k) relative to observations (n). 'Big' data is used to describe settings where you have more regressors than observations.

It turns out that you can write (see CEF Material),

$$y = m(\mathbf{x}) + u$$

where u has the nice property $E[u|\mathbf{x}] = 0$.

If you **assume** that $m(\mathbf{x})$ is linear (in parameters), then the best predictor of y is

$$E[y|\mathbf{x}] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$$

and,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

[3] Referred to as the population regression function.

Population Regression Function

The function $m(\mathbf{x})$ is called the population regression function and $\mathbf{x}\beta$ is referred to as the linear population regression function (or linear projection). It may be that $m(\mathbf{x}) = \mathbf{x}\beta$.

Approach #3: Economics

The traditional approach to Econometrics is rooted in the formal economic modelling. A famous example is the Solow Growth model:

Step 1: Specify production function

Popular choice is Cobb-Douglas

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$$

Step 2: Express in terms of per-worker variables:

$$y_t = A_t k_t^{\alpha}$$

where $y_t = Y_t/L_t$ and $k_t = K_t/L_t$.

Step 3: Log linearize

$$\ln y_t = \ln A_t + \alpha \ln k_t$$

Now we have a linear model with a slope parameter α - the parameter of interest.

BUT we have a problem:

1. A_t - which represents TFP - is not observed in the data
2. There is no error term! The model is deterministic.

Missing error term

Regression models have inherent uncertainty: the error term. Because of the error term, the realized value of y need not match the value predicted by the population regression function (e.g. $\mathbf{x}\beta$). However, many economic models have no inherent uncertainty. And when they do, this uncertainty may be of a different form to that observed in real economic

variables. Indeed, this issue of 'missing error term' is a very old problem in Economics. As Morgan (1992) describes, economists were initially resistant to the adoption statistical methods. The hesitancy to consider these methodologies was down to a belief that the 'laws' studied by economists acted like laws of nature. That is, they could be directly measured, just as you would measure gravitational force. The was therefore no inherent uncertainty in the model. Indeed, early attempts to rationalize uncertainty in economic data often turned to measurement error. The model didn't fit the data perfectly because the variables were measured with error.

Step 4: Make an assumption

Assume that TFP follows a particular time-series process (i.e. DGP):[4]

$$\ln A_t = \ln A_0 + gt + u_t$$

where gt is a deterministic trend and u_t a stochastic shock.

NOW we have a model of the form

$$\ln y_t = \ln A_0 + gt + \alpha \ln k_t + u_t$$

which we can rewrite as a linear regression model with 3 unknown parameters:

$$\ln y_t = \beta_0 + \beta_1 t + \beta_2 \ln k_t + u_t$$

Mankiw, Romer & Weil (1992)

The above equation is a time-series regression model that requires rather strong assumptions. Mankiw, Romer, and Weil (1992) derives a more complete approach, based on a model with inherent uncertainty. The paper uses a linear approximation of dynamics around the steady state and derives an equation which can be estimated using a cross-section of country-level data. This approach is discussed by Prof. Daron Acemoglu in his comprehensive lecture notes.

[4] If you are keen to learn more about these approaches you should consider taking EC4437 *Applied Macroeconomics*.

Approach #4: Causal Inference

A more modern approach, popularized by applied microeconomists, is not rooted in formal economic modelling. Instead, it focuses on a set of research designs (methodologies) that target the causal relationship between two variables. These are often taught as *Microeconometrics*.[5]

> **Microeconometrics**
>
> Microeconometrics focuses on the identification and estimation of causal relationships. It builds on the Neyman-Rubin Causal Model which includes the Potential Outcomes Framework. Related to this field are literatures on randomized experiments, causal inference, treatment effects, and policy evaluation. And these methods are prolific across applied microeconomics fields; including, Labour, Health, Education, Development, Public Policy, etc.
> If you are interested in these topics, you should consider reading:
>
> 1. Angrist and Pischke (2014) *Mastering 'Metrics* (good introduction)
> 2. Angrist and Pischke (2009) *Mostly Harmless Econometrics* (more advanced notation)
> 3. Cunningham (2021) *Causal Inference: The Mixtape* (up to date and available online)

For example,

$$\text{Marginal Tax Rate} \rightarrow \text{Labour Supply}$$

A researcher might choose to estimate this relationship within a linear regression model framework, by specifying the *estimating equation*:

$$hrswrk_i = \beta_0 + \beta_1 \text{MTR}_i + u_i$$

The *estimate* of β_1 tells us about how a change in MTR affects $hrswrk$ in the data.

BUT if we were to estimate this relationship in the data, do we really think it will tell us something causal? Consider:

- those who face a higher marginal tax rate do so because they earn more;
- and those with higher income might be people with higher education;

[5] If you are keen to learn more about these approaches you should consider taking EC4425 *Econometrics of Impact Evaluation*.

- …, etc.

Can we account for these differences in the model?

$$hrswrk_i = \beta_0 + \beta_1 MTR_i + \beta_2 income_i + \beta_3 yrsedu_i + u_i$$

This is where we get into the idea of adding additional regressors as "control" variables.

Classical Linear Regression Model

What constitutes a model? In some sense, a model is just a collection of assumptions. Here are the assumptions that govern the CLRM. Some of them will be more relevant once we discuss the estimation of the CLRM. We will follow Wooldridge's notation of labelling these MLR. 1-6. In Chapter 2, Wooldridge defines SLR. 1-6. These are a special case of MLR. 1-6 for a model with a simple linear model (i.e. single regressor and intercept).

> Alternatives
>
> Please be aware that there alternative ways to specify the CLRM assumptions. One noteable variant, found in a number of texts (and websites), considers the case of non-random regressors. That is, the value of regressors does not change with sampling.

MLR #1

These assumptions have the abbreviation MLR for Multiple Linear Regression.

MLR-1: Linear in parameters

The population model is given by,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

It may seem obvious, but it is nonetheless an assumption that the model is **correctly specified** as linear in parameters.

MLR #2

MLR-2: Random sampling

Random sample of n observations, $\{\mathbf{x}_i, y_i\} : \ i = 1, \ldots, n$, drawn from the population model

Random sampling is sometimes referred to as *independently and identically distributed* (iid).

- The 'identically distributed' part can be relaxed (i.e. error variance)

Together with MLR-1, it gives us

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + u_i \qquad i = 1, \ldots, n$$

- Needed for Law of Large Numbers and Central Limit Theorem
- Sampling is the primary source of uncertainty the model.

MLR #3

MLR-3: No perfect collinearity

No *exact* linear relationship among regressors (including the constant).

This assumption is key for *identification*. It ensures that each parameter is uniquely identified.

Example

Consider a regression of hours-worked against temperature, measured in both F° and C°,

$$hrswrk_i = \beta_0 + \beta_1 tempC_i + \beta_2 tempF_i + u_i$$

The model has 3 parameters.

BUT there is a linear formula that links temperature in $F^\circ \Rightarrow C^\circ$: $F^\circ = 9/5 \cdot C^\circ + 32$.

- Plug this into the equation:

$$\begin{aligned} hrswrk_i =& \beta_0 + \beta_1 tempC_i + \beta_2(9/5 \cdot tempF_i + 32) + u_i \\ =& (\beta_0 + \beta_2 \cdot 32) + (\beta_1 + \beta_2 \cdot 9/5) tempC_i + u_i \\ =& \gamma_0 + \gamma_1 tempC_i + u_i \end{aligned}$$

The model has 2 parameters now.

- The third was not identified because the variable $tempF_i$ is perfectly collinear with $tempC_i$ and the constant in the model.

MLR #4

MLR-4: Zero conditional mean

The error term has expected value of 0 given the any value of the explanatory variables

$$E[u|\mathbf{x}] = 0$$

With random sampling, then MLR-4 $\Rightarrow E[u_i|\mathbf{x}_i]$ for $i = 1, \dots, n$.

- weaker than full independence: $u \perp \mathbf{x}$;
- but stronger than uncorrelatedness: $Cov(\mathbf{x}, u) = 0$

We can also show the following:

1. MLR-4 implies that the unconditional mean of the error term is 0 (Law of Iterated Expectations; see CEF notes).

$$E[u|\mathbf{x}] = 0 \Rightarrow E[u] = 0$$

2. MLR-4 implies the error term is uncorrelated with all regressors

$$E[u|\mathbf{x}] = 0 \Rightarrow E[ux_j] = Cov(u, x_j) = 0 \quad j = 1, 2, \dots, k$$

3. MLR-4 implies that we can *treat* $\mathbf{x}$ as fixed across repeated samples (even if they are not; i.e. also random variables).

    - If x is fixed, there is only source of uncertainty: u.
    - Allows us to treat x as non-random in derivations; e.g. variance of estimator.

MLR #5

MLR-5: Homoskedasticity

The variance of the error term is the same given any value of $\mathbf{x}$.

$$Var(u|\mathbf{x}) = \sigma^2$$

With random sampling, then MLR-5 $\Rightarrow Var(u_i|\mathbf{x}_i) = \sigma^2$ for $i = 1, \dots, n$.

- Needed to define the variance of the estimator
- Needed for the famous BLUE property of OLS (Gauss Markov Theorem)
- It can be relaxed; such errors are referred to as heteroskedastic

MLR #6

MLR-6: Normality

The conditional distribution of the error term is normal.

$$u|\mathbf{x} \sim N(0, \sigma^2)$$

- Needed to know the *finite* distribution of the estimator
- Can be relaxed, but then only asymptotic distribution is known

Simple case

SLR model

Let's consider the simplest case: a bivariate regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Under MLR. 1-4 (or Wooldridge's SLR. 1-4):

$$\begin{aligned} E[y_i|x_i] =& E[\beta_0 + \beta_1 x_i + u_i|x_i] \\ =& E[\beta_0|x_i] + E[\beta_1 x_i|x_i] + E[u_i|x_i] \\ =& \beta_0 + \beta_1 x_i + E[u_i|x_i] \\ =& \beta_0 + \beta_1 x_i \end{aligned}$$

Identification

Recall, under MLR-4

1. $E[u_i] = 0$
2. $E[u_i x_i] = 0$

We can use these two moments to demonstrate the identification of β_0 and β_1.

Step 1: Substitute $u_i = y_i - \beta_0 - \beta_1 x_i$ into (1)

$$0 = E[u_i] = E[y_i - \beta_0 - \beta_1 x_i] \Rightarrow \beta_0 = E[y_i] - \beta_1 E[x_i]$$

Step 2: Substitute $u_i = y_i - \beta_0 - \beta_1 x_i$ into (2)

$$0 = E[u_i x_i] = E[(y_i - \beta_0 - \beta_1 x_i)x_i] = E[y_i x_i] - \beta_0 E[x_i] - \beta_1 E[x_i^2]$$

Step 3: Substitute the solution for β_0

$$0 = E[y_i x_i] - \beta_0 E[x_i] - \beta_1 E[x_i^2] = E[y_i x_i] - (E[y_i] - \beta_1 E[x_i])E[x_i] - \beta_1 E[x_i^2]$$

Manipulate and solve

$$0 = \underbrace{E[y_i x_i] - E[y_i]E[x_i]}_{Cov(y_i, x_i)} - \beta_1 (\underbrace{E[x_i^2] - E[x_i]^2}_{Var(x_i)})$$

Arrive at the neat solution:

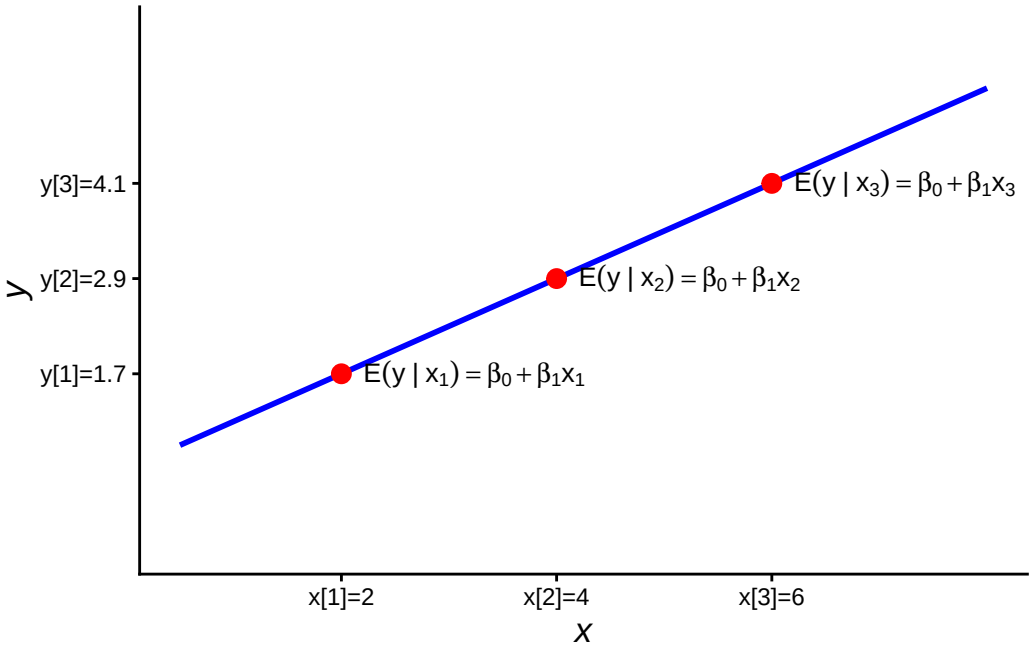
$$\beta_1 = \frac{Cov(y_i, x_i)}{Var(x_i)}$$

and,

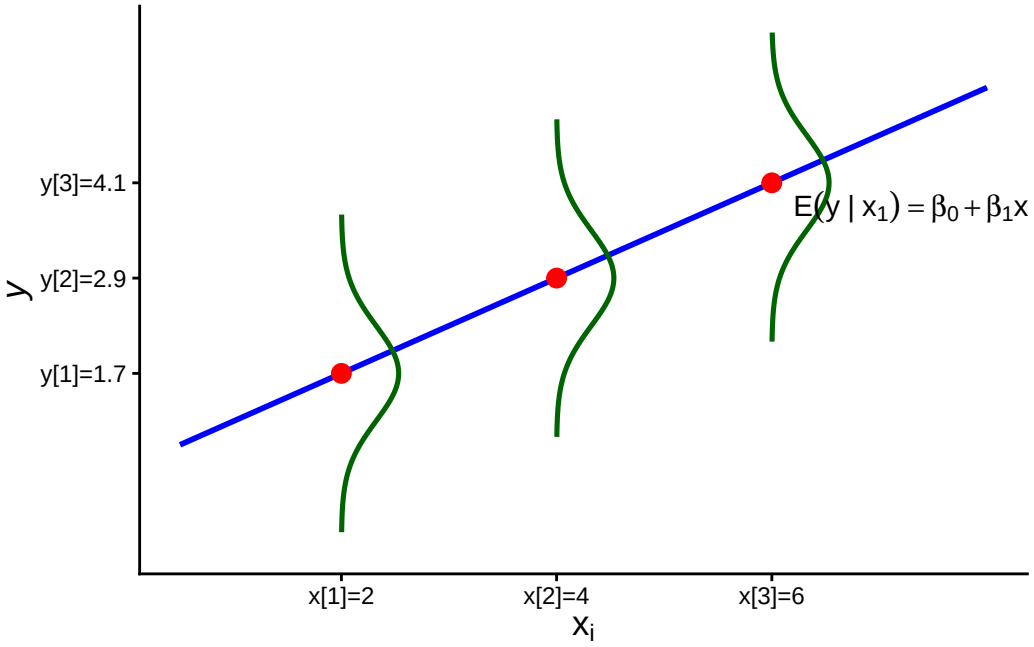
$$\beta_0 = E[y_i] - \beta_1 E[x_i]$$

Both population parameters can be written as expressions of "observable" moments in the data. In this sense, they are *identified*.

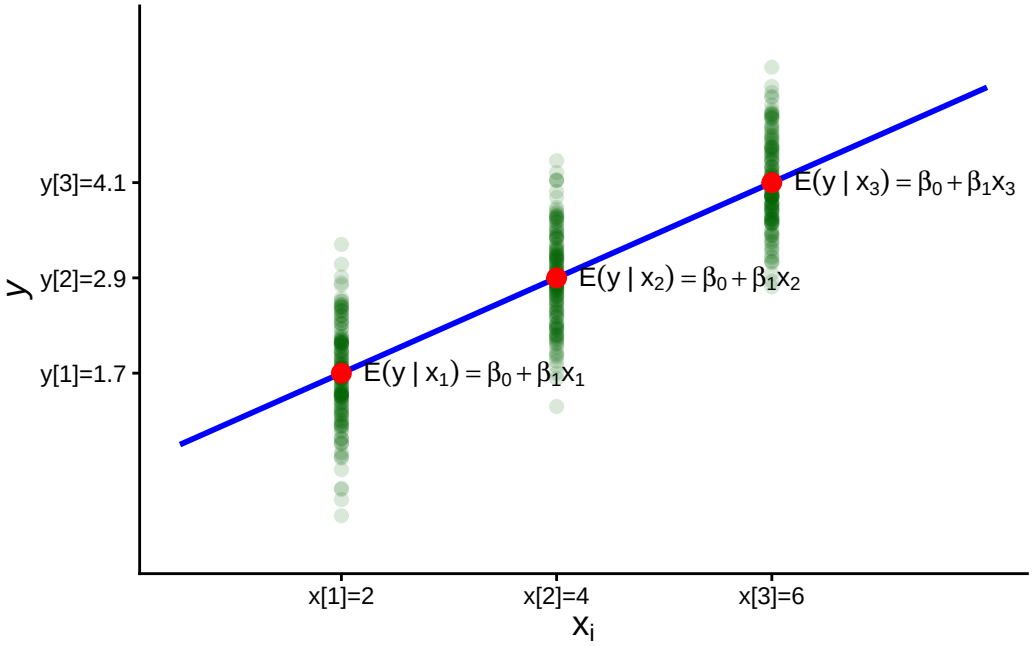
Let's plot $E[y|x] = 0.5 + 0.6x$, the (linear) population regression function.



Around the $E[y_i|x_i]$, the error term is normaly distributed: $u_i \sim N(0, 0.6^2)$



This is what it would look like if you sampled $n = 50$ errors for x from that distribution.



For a given x, more of the observations will be concentrated around the *conditional* mean.

Interpretation: Continuous regressor

This then gives us a clear interpretation of β_0 and β_1

$$\begin{aligned}\beta_0 =& E[y_i|x_i = 0] \\ \beta_1 =& \frac{dE[y_i|x_i]}{dx_i}\end{aligned}$$

Note, β_0 may not have a "realistic" interpretation.

- e.g., if y is level of exports (to US) and x is USD exchange rate of country i.

> **Warning**
>
> The interpretation of β_1 as a derivative assumes that x_i is a continuous variable.

Why not?

$$\beta_1 = \frac{dy_i}{dx_i}$$

- Without MLR. 1-4 (zero condition mean), we can't rule out a relationship bewteen u and x
  - e.g., u could include powers of x: $u_i = \beta_2 x_i^2 + v_i$
- With MLR. 1-4, $E[y_i|x_i] = \beta_0 + \beta_1 x_i$ (i.e. the population regression function)

Interpretation: Binary regressor

What if $x \in \{0, 1\}$?

- We cannot differentiate $E[y_i|x_i]$.

Solution: difference

$$\begin{aligned} E[y_i|x_i = 0] =& \beta_0 + \beta_1 \times 0 = \beta_0 \\ E[y_i|x_i = 1] =& \beta_0 + \beta_1 \times 1 = \beta_0 + \beta_1 \end{aligned}$$

$$\Rightarrow \beta_1 = E[y_i|x_i = 1] - E[y_i|x_i = 0]$$

Multivariate case

MLR model

In the multivariate case we have models of the form ($k = 3$),

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + u_i$$

As with the SLR case, under MLR 1-4:

$$E[y_i|\mathbf{x}_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}$$

Interpretation: Continuous regressor (linear)

Suppose, the linear population regression function is *also* linear in x.[6]

We can now think of each slope coefficient as

$$\beta_j = \frac{\partial E[y_i|\mathbf{x}_i]}{\partial x_{ij}} \quad j = 1, 2, 3$$

This is a *partial derivative*.

- Change in the conditional mean of y for a 1 unit change in regressor x_j, **holding x_2 and x_3 fixed**
- Why additional regressors in a model are referred to as "control variables"; their presence changes the interpretation of the slope coefficient.

Differential

The mean of the outcome will change in the following manner with the regressors:

$$\Delta E[y_i|\mathbf{x}_i] = \frac{\partial E[y_i|\mathbf{x}_i]}{\partial x_{i1}}\Delta x_{i1} + \frac{\partial E[y_i|\mathbf{x}_i]}{\partial x_{i2}}\Delta x_{i2} + \frac{\partial E[y_i|\mathbf{x}_i]}{\partial x_{i3}}\Delta x_{i3}$$

So, if you ask what is the change in (expected value of y) from a 10-unit change in x_1:

$$\Delta E[y_i|\mathbf{x}_i] = \beta_1 \times 10$$

Interpretation: Continuous regressor (polynomial)

Suppose, the *linear* population regression function includes higher-order polynomials of some x's.[7]

$$E[y_i|\mathbf{x}_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \beta_3 x_{i3}$$

The derivative (w.r.t. x_1) now depends on the value of x_1:

$$\frac{\partial E[y_i|\mathbf{x}_i]}{\partial x_{i1}} = \beta_1 + 2\beta_2 x_{i1}$$

[6] And additively separable, which rules out interactions between variables.
[7] And additively separable, which rules out interactions between variables.

NBER example

From Berg, Curtis, and Mark (2026, 6) "Temperature Fluctuations and Economic Conditions: Evidence from Weekly U.S. Data"

$$\sum_{j=0}^{h} y_{i,t+j} = \beta_{1,h}\tau_{i,t} + \beta_{2,h}\tau_{i,t}^2 + \delta_h \tau_{i,t-1} + \sum_{j=1}^{2} \gamma_{h,j} y_{i,t-j} + \alpha_i + \theta_t + \epsilon_{i,t+h}, \tag{1}$$

- $\sum_{j=0}^{h} y_{i,t+j}$ $y_{i,t}$ - is the Economic Conditions Index for state i in week t
- $\tau_{i,t}$ - seasonally adjusted temperature

Interpretation: Continuous regressor (interactions)

Suppose, the *linear* population regression function includes higher-order polynomials of some x, but had some interactions.

$$E[y_i|\mathbf{x}_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1} \cdot x_{i3} + \beta_3 x_{i3}$$

The derivative (w.r.t. x_1) now depends on the value of x_3:

$$\frac{\partial E[y_i|\mathbf{x}_i]}{\partial x_{i1}} = \beta_1 + \beta_2 x_{i3}$$

NBER example

From Kristian Blickle and Ma (2025, 10) "The Dynamics of Deposit Flightiness and its Impact on Financial Stability"

$$Flow_{jt} = \beta_1 \widehat{DepRate}_{jt} + \beta_2 \widehat{DepRate_{jt} \cdot AggFlow_t} + FE_t + FE_j + Controls + \epsilon_{jt}, \tag{3.2}$$

- $Flow_{jt}$ - deposit flow of bank j in quarter t
- $DepRate_{jt}$ - deposit rates for bank j in quarter t
- $AggFlow_t$ - one-year cumulative deposit flow in quarter t

Interpretation: binary variable

Suppose, that $x_1 \in \{0, 1\}$.

$$E[y_i|x_{i1} = 0, x_{i2}, x_{i3}] = \beta_0 + 0 + \beta_2 x_{i2} + \beta_3 x_{i3}$$

and,

$$E[y_i|x_{i1} = 1, x_{i2}, x_{i3}] = \beta_0 + \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3}$$

Thus,

$$\beta_1 = E[y_i|x_{i1} = 1, x_{i2}, x_{i3}] - E[y_i|x_{i1} = 0, x_{i2}, x_{i3}]$$

NBER example

From Karlan, Rigol, and Roth (2026, 24) "When Microenterprises Grow, Are Consumers Better Off? Evidence from Large Loans to Microenterprises in Chile"

$$y_i = \alpha + \beta T_i + X_i \delta + \varepsilon_i \tag{2}$$

- T_i - indicator for treatment (loan receipt): treated = 1, control = 0
- X_i - firm characteristics

Log transformation

It is common in economics to apply a log-transformation to economic variables.

$$\ln(y)$$

WHy?

- helps to deal with outliers
- some variables are (close to) log-normal
- log-linearization is used to transform economic models
- changes the interpretation; e.g. growth rates of macroeconomic variables

Interpretation: log-level

Suppose,

$$E[\ln(y_i)|\mathbf{x}_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}$$

$$\beta_j = \frac{\partial E[\ln(y_i)|\mathbf{x}_i]}{\partial x_{ij}}$$

The coefficient is therefore measured in log-units of y.

The relation to a change in the (expected) level of y is given by,

$$\%\Delta\ E[y_i|\mathbf{x}_i] = (exp(\beta_j) - 1) \times 100$$

For reasonably small values of β_1 (i.e. within the range $[-0.1, 0.1]$) this can be approximated by,

$$\%\Delta\ E[y_i|\mathbf{x}_i] \approx \beta_j \times 100$$

A 1-unit change in x_{ij} is associated with a $\approx \beta_j \times 100$ **percentage** change in the expected value of y.

This referred to as a **semi-elasticity**.

NBER example

From Eriksson, Lin, and Niemesh (2026, 16) "Immigration and Child Mortality during the Age of Mass Migration"

$$ln(M_{it}) = \alpha + \beta SF_{it} + \eta_i + \sigma_t + (\gamma_{c(i)t} \times \sigma_t) + \epsilon_{it} \tag{4}$$

- $\ln(M_{it})$ - log of the mortality rate in the town-year
- SF_{it} - share of the town's population that was foreign-born

Interpretation: level-log

Suppose,

$$E[y_i|\mathbf{x}_i] = \beta_0 + \beta_1 \ln(x_{i1}) + \beta_2 x_{i2} + \beta_3 x_{i3}$$

Then,

$$\beta_1 = \frac{\partial E[y_i|\mathbf{x}_i]}{\partial \ln(x_{i1})}$$

The coefficient is measured in y.

A 1 **percent** increase in x is exactly $x \times 1.01$. This is equivalent to a change in $\ln(x)$ of,

$$\ln(x_i \times 1.01) - \ln(x_i) = \ln(1.01) \approx 0.01$$

Thus, a 1 **percent** increase in (the level of) x is associated with a $\beta_1 \times 0.01 = \beta_1/100$ increase in the expected value of y. Or, more accurately

$$\Delta E[y_i|x_i] = \beta_1 \times \ln(1.01)$$

This is also a **semi-elasticity**.

NBER example

From Cordova and Schaller (2026, 23) "Local Labor Demand and Achievement Gaps: Evidence from the Great Recession"

$$Y_{cgt} \;=\; \beta\, \Delta \ln E_{ct} + \alpha_c + \delta_t + \gamma_g + \varepsilon_{cgt}, \tag{2}$$

- Y_{cgt} - cohort-standardized achievement in county c, grade g, year t
- $lnEct$ - log of employment in country c, year t (ignore the first-difference Δ)

Interpretation: log-log

Suppose,

$$E[\ln(y_i)|\mathbf{x}_i] = \beta_0 + \beta_1 \ln(x_{i1}) + \beta_2 x_{i2} + \beta_3 x_{i3}$$

Then,

$$\beta_1 = \frac{\partial E[\ln(y_i)|\mathbf{x}_i]}{\partial \ln(x_{i1})}$$

This is an **elasticity** measure: β_1 is the % change in the expected value of y from a 1 % change in x

NBER example

From Bai, Bian, and Jia (2026, 9) "Politics-driven Market Access and Its Cost: Evidence from China's Grand Canal"

$$\ln MA_{it} = \alpha \ln MA_{it}^{opt,pop_0} + \sum_{t=1}^{t=9} (Z_i \times I_{\rho=t})\gamma^{\rho} + \lambda_i + \eta_t + \epsilon_{it}. \tag{3}$$

- $\ln MA_{it}$ - (log of) market access in grid i, year t
- $\ln MA_{it}^{opt,pop_0}$ - (log of) optimal market access based on historic constraints (e.g. Tang-dynasty population)

Q&A

Questions

1. How many parameters are there in the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$? And how many regressors?
2. TRUE/FALSE: are the following models linear in parameters?
   2.1. $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$
   2.2. $y = \theta_0 + \theta_1 x + \theta_1^2 z + u$
   2.3. $y = \alpha + \beta x + \gamma x^2 + u$

2.4. $y = \alpha + \beta x + \mathbf{z}\gamma + u$

2.5. $y = exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + u)$

3. Suppose, $exp_i = age_i - (edu_i + 6)$. Is the following model identified? What if you excluded age_i from the model?

$$\ln(wage_i) = \beta_0 + \beta_1 exp_i + \beta_2 age_i + \beta_3 edu_i + u_i$$

Answers

1. $k + 1$ parameters, k regressors
2. TRUE/FALSE: are the following models linear in parameters?

   2.1. TRUE

   2.2. FALSE

   2.3. TRUE

   2.4. TRUE

   2.5. FALSE

3. No, since exp_i can be perfectly explained by the other variables (including the intercept). If age_i is removed, then the model is identified.

References

Bibliography

Angrist, Joshua D, and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton university press.

———. 2014. *Mastering'metrics: The Path from Cause to Effect*. Princeton university press.

Bai, Ying, Xiaoyu Bian, and Ruixue Jia. 2026. "Politics-Driven Market Access and Its Cost: Evidence from China's Grand Canal." Working Paper 35721. Working Paper Series. National Bureau of Economic Research. https://doi.org/10.3386/w35721.

Berg, Kimberly A., Chadwick C. Curtis, and Nelson C. Mark. 2026. "Temperature Fluctuations and Economic Conditions: Evidence from Weekly u.s. Data." Working Paper 35529. Working Paper Series. National Bureau of Economic Research. https://doi.org/10.3386/w35529.

Cordova, Karla, and Jessamyn Schaller. 2026. "Local Labor Demand and Achievement Gaps: Evidence from the Great Recession." Working Paper 35639. Working Paper Series. National Bureau of Economic Research. https://doi.org/10.3386/w35639.

Cunningham, Scott. 2021. *Causal Inference: The Mixtape*. Yale university press.

Eriksson, Katherine, Peter Zhixian Lin, and Gregory Niemesh. 2026. "Immigration and Child Mortality During the Age of Mass Migration." Working Paper 35743. Working Paper Series. National Bureau of Economic Research. https://doi.org/10.3386/w35743.

Karlan, Dean, Natalia Rigol, and Benjamin N. Roth. 2026. "When Microenterprises Grow, Are Consumers Better Off? Evidence from Large Loans to Microenterprises in Chile." Working Paper 35729. Working Paper Series. National Bureau of Economic Research. https://doi.org/10.3386/w35729.

Kristian Blickle, Xu Lu, Jian Li, and Yiming Ma. 2025. "The Dynamics of Deposit Flightiness and Its Impact on Financial Stability." Working Paper 34128. Working Paper Series. National Bureau of Economic Research. https://doi.org/10.3386/w34128.

Mankiw, N Gregory, David Romer, and David N Weil. 1992. "A Contribution to the Empirics of Economic Growth." *The Quarterly Journal of Economics* 107 (2): 407–37.

Morgan, Mary S. 1992. *The History of Econometric Ideas*. Cambridge university press.

Wooldridge, Jeffrey M. 2025. "Introductory Econometrics: A Modern Approach."